

AUTOMATIC EVENING LIGHT With Cut-Off Timer

PRADEEP VASUDEVA

Designing a circuit to switch on a light at dusk and off at dawn is straightforward, and many such designs have been featured in EFY. However, lights controlled this way remain on all night, resulting in wasted electricity and reduced lamp lifespan. In most homes and business premises, outdoor lights are turned on at dusk and switched off after a few hours. The device described here automates this process, activating the light in the evening and switching it off after 2 to 6 hours. This approach is particularly effective for securing locked homes, as the timed lighting creates the illusion of occupancy and deters potential intruders.

Circuit and working

Fig. 2 shows the circuit diagram of the automatic evening light. It is built around the dual op-amp IC LM358. The first op-amp, built around IC1A, is used as a comparator. The inverting pin 2 of the comparator is biased at half the supply voltage by resistors R1 and R2. During the daytime, the resistance of the LDR connected to the non-inverting pin 3 of the comparator is low. The voltage at the non-inverting pin 3 is lower than the voltage at the inverting pin 2, so the output at pin 1 of the comparator is low.

In the evening, the resistance of the LDR increases and becomes more than the resistance of R3 plus VR1, and the output at pin 1 of the comparator becomes high (that is, about 1.5 volts below the supply voltage). Potentiometer VR1 serves

as sensitivity control.

The second op-amp is built around IC1B of IC LM358. Its inverting input pin 6 is connected to supercapacitor C1. Resistor R4 and LED1 drop the output of comparator IC1A at pin 1 to the forward voltage of LED1, which is about 2V. This voltage is used to charge the supercapacitor through resistor R6. This protects the supercapacitor from damage due to high voltage and ensures that it is charged at a constant rate, irrespective of supply voltage fluctuations.

The non-inverting input pin 5 of the second comparator IC1B is connected to the forward voltage of LED2 through potentiometer VR2. LED2 also acts as a power-on indicator. Initially, the supercapacitor C1

is discharged, and the voltage at the non-inverting input pin 5 is higher than the voltage at the inverting input pin 6, so the output at pin 7 of the second comparator IC1B is high.

Thus, in the evening, the outputs of both comparators IC1A and IC1B are high. The outputs of both comparators are applied to transistor T1 through diodes D1 and D2 and pull-up resistor R8, which function as an AND gate. This conducts the transistor T1 to switch on the light through the contacts of relay RL1.

After a few hours, the voltage at supercapacitor C1 exceeds the voltage applied to the non-inverting pin 5 of the second comparator, and the output at pin 7 becomes low. This cuts off transistor T1 and switches off the light. The output of the first com-

parator remains high and LED1 continues to glow, and C1 continues to charge.

In the morning, the output of the first comparator also becomes low and LED1 turns off. Super capacitor C1

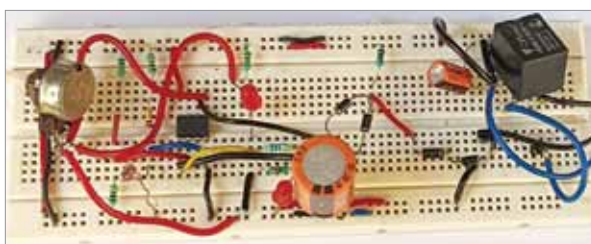


Fig. 1: Prototype on breadboard

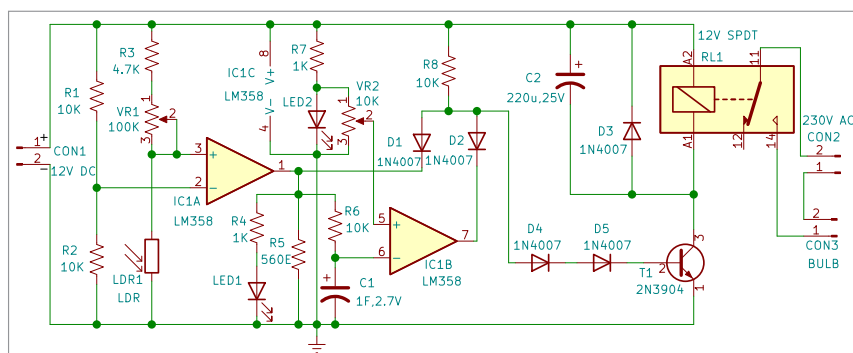


Fig. 2: Circuit diagram